Genetics Problems Assignment

1. In humans, brown eyes (B) are dominant to blue eyes (b). A homozygous brown eyed parent has a child with a blue eyed parent. Determine the possible genotypes and phenotypes of the offspring expected.

2. In mice, yellow hair (H) is dominant to normal hair colour (h). Show the expected genotypes and phenotypes that would result from a cross between two heterozygous yellow mice.

3. In peas, round peas (R) are dominant to wrinkled peas(r). If a hybrid round pea is crossed with a wrinkled pea, explain the likely results of the cross.

4. In mice, black coat colour (B) is dominant to brown (b). What are the possible genotypes and phenotypes when a hybrid black mouse is crossed with another hybrid black mouse?

5. In guinea pigs, rough coat (R) is dominant over smooth coat (r). What are the possible genotypes and phenotypes of a cross between a pure breed smooth coated guinea pig and a hybrid rough coated guinea pig?

6. Albino rabbits have white hair and pink eyes. The gene for albinism is recessive. If a pure bred normal rabbit is crossed with an albino one, what are the different genotypes and phenotypes that could result and their frequencies.

7. For humans, blond hair is recessive to dark hair. What are the possible genotypes and phenotypes for hair colour that could appear in the children of a father who is pure breed for dark hair and a mother who is hybrid for dark hair?

8. What genotypes and phenotypes will be produced in the offspring of a hybrid black female mouse and a white male mouse, given that black is dominant to white.

9. If a pure breeding male black guinea pig is mated with a pure breeding female white guinea pig, what possible offspring could be produced, and in what proportions, given that black coat colour is dominant over white.

10. In cattle, black coat is dominant to red coat. If a pure breed red cow is mated with a hybrid black bull what are the frequencies of genotype and phenotype for coat colour in the offspring produced.

11. In rabbits, black fur is dominant to brown fur. Two black hybrid rabbits are crossed. Show the expected genotypes and phenotypes of their offspring.

12. In mice, the gene for straight tail is dominant to curly tail.
    a. Find the genotypes and phenotypes produced when a pure bred mouse with a straight tail is crossed with a pure bred mouse with a curly tail. This is the first generation.

    b. Find the genotypes and phenotypes produced if two mice from the first generation are mated.

13. In fruit flies, the normal wing shape is dominant to the 'dumpy' wing shape.
    a. Two heterozygous flies are crossed. Calculate the expected proportion of both genotype and phenotype.

    b. If the above cross resulted in 240 offspring, how many of each phenotype would be expected?

    c. How many of each genotype would you expect out of the 240 offspring?

14. In human beings, brown eyes are dominant over blue eyes. Suppose a blue eyed man marries a brown eyed woman whose father was blue eyed. What are the expected genotypes and phenotypes of their offspring?

15. In tomatoes red fruit is dominant to yellow fruit. A farmer has a red fruiting plant and doesn't know if it is a hybrid or a pure-bred. To find out, he crosses it with a yellow plant.
    a. If the red tomato is a hybrid, what sort of fruit will the offspring have?

    b. If the red tomato is a pure-bred, what sort of fruit will the offspring have?

16. A brown eyed man, Bill, married a brown eyed woman, Jill. They have a blue eyed daughter and a brown eyed son. What are all the possible genotypes of the people mentioned, given that brown eyes is dominant to blue eyes?